# Assignment #1

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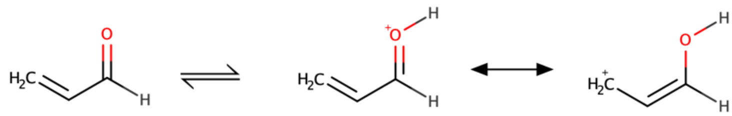
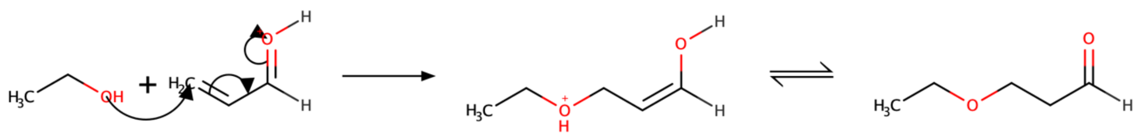
I was asked to answer question 2F from Appendix 5 of the textbook.

This reaction was an acid-catalyzed addition reaction of ethanol to 3-butenal (AKA acrolein).

One possible mechanism begins with the equilibrium protonation of the aldehyde carbonyl group. The positive charge on the oxygen is shared with the β-carbon.



A nucleophile may add either at the carbonyl carbon or the β-carbon. In this case we observe addition at the β-carbon. The immediate addition product will be a protonated ether. This will give the acidic proton back to the solvent system. The acid is returned as expected in an acid-catalyzed reaction. In acidic conditions the enol will rapidly tautomerize to the keto form giving 3-ethoxypropanal.



This is like the Michael reaction that I learned about in organic chemistry, except that it occurs in acidic rather than basic conditions. It's amazing that previous courses are starting to become useful to me.

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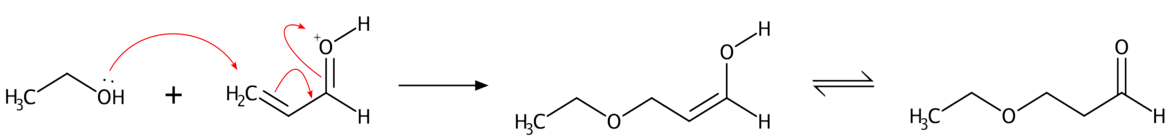
Tip: *I can improve the chemical reaction diagram by pasting the molecules in the diagram above into LibreOffice Draw and then using the draw tools to make better arrows. Consider the diagram below*



[1]This is an example assignment to demonstrate what I am looking for. Observe the chemical reaction schemes that use arrows to denote electron flow and the brief explanatory text. The chemical diagrams were both at the same scale. This is important in a document as different sizes could be interpreted as different emphasis or carelessness. *LibreOffice* was used to create this document. I used *MarvinSketch* to make the diagrams. The arrows in *MarvinSketch* are not very stylish, but I will never blame you for the quality of a free tool.